

Universidad Tecnológica de Emiliano Zapata (UTEZ)

Carrera: **Ingeniería en desarrollo y gestión de
software**

Materia: **Extracción de conocimiento de Bases
de Datos**

Alumno: **Ruiz Santos Orlando**

Profesor: **Eliel Rodriguez**

Examen U2

Grado: **9º**

Grupo: **A**

Fecha: **05/07/26**

1. Captura del repositorio con la estructura solicitada:

The screenshot shows a web browser window displaying a GitHub repository. The browser's address bar shows the URL `github.com/RuizOr/Unidad2_ETL_Biblioteca_RuizSantosOrlando`. The repository name is `Unidad2_ETL_Biblioteca_RuizSantosOrlando` and it is marked as `Public`. The repository has 0 forks, 0 stars, and 0 watchers. The main branch is `main`, and there is 1 branch and 0 tags. The repository contains a commit by `RuizOr` with the message `feat: tarea terminada`, dated 3 minutes ago. The commit includes a table of files and folders:

File/Folder	Commit Message	Time Ago
<code>data</code>	<code>feat: estructura del proyecto</code>	1 hour ago
<code>evidencias</code>	<code>feat: tarea terminada</code>	3 minutes ago
<code>scripts</code>	<code>feat: tarea terminada</code>	3 minutes ago
<code>sql</code>	<code>feat: tarea terminada</code>	3 minutes ago
<code>README.md</code>	<code>feat: tarea terminada</code>	3 minutes ago

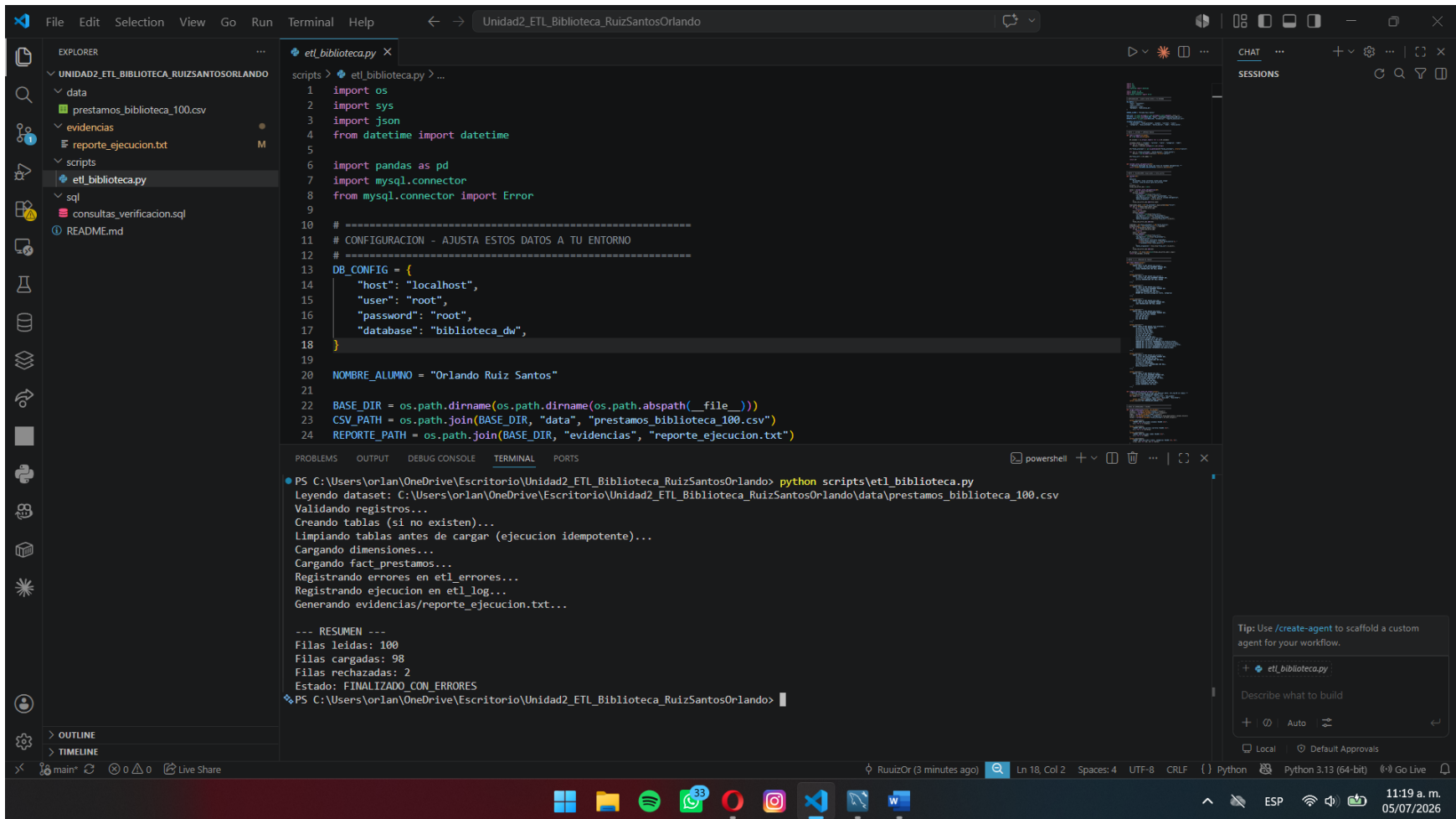
The repository also has a `README` file. The `README` content is as follows:

Unidad 2 - ETL Prestamos de Biblioteca

1. Objetivo del proyecto

The right sidebar of the repository page shows the `About` section, which states: `No description, website, or topics provided.` Below this, there are links for `Readme`, `Activity`, `0 stars`, `0 watching`, and `0 forks`. The `Releases` section shows `No releases published` and a link to `Create a new release`. The `Packages` section shows `No packages published` and a link to `Publish your first package`. The `Contributors` section shows `1` contributor.

2. Captura de ejecución del script:



The screenshot displays the Visual Studio Code interface with a Python script named `etl_biblioteca.py` open in the editor. The script is located in the `scripts` directory of a project named `UNIDAD2_ETL_BIBLIOTECA_RUIZSANTOSORLANDO`. The script's content is as follows:

```
1 import os
2 import sys
3 import json
4 from datetime import datetime
5
6 import pandas as pd
7 import mysql.connector
8 from mysql.connector import Error
9
10 # =====
11 # CONFIGURACION - AJUSTA ESTOS DATOS A TU ENTORNO
12 # =====
13 DB_CONFIG = {
14     "host": "localhost",
15     "user": "root",
16     "password": "root",
17     "database": "biblioteca_dw",
18 }
19
20 NOMBRE_ALUMNO = "Orlando Ruiz Santos"
21
22 BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
23 CSV_PATH = os.path.join(BASE_DIR, "data", "prestamos_biblioteca_100.csv")
24 REPORTE_PATH = os.path.join(BASE_DIR, "evidencias", "reporte_ejecucion.txt")
```

The terminal window shows the execution of the script using the command `python scripts\etl_biblioteca.py`. The output indicates that the dataset `C:\Users\orlan\OneDrive\Escritorio\Unidad2_ETL_Biblioteca_RuizSantosOrlando\data\prestamos_biblioteca_100.csv` was read, records were validated, tables were created (if they didn't exist), and data was loaded. The script also generated a report in `evidencias\reporte_ejecucion.txt`. The summary shows 100 files read, 98 files loaded, and 2 files rejected. The final status is `FINALIZADO CON ERRORES`.

```
PS C:\Users\orlan\OneDrive\Escritorio\Unidad2_ETL_Biblioteca_RuizSantosOrlando> python scripts\etl_biblioteca.py
Leyendo dataset: C:\Users\orlan\OneDrive\Escritorio\Unidad2_ETL_Biblioteca_RuizSantosOrlando\data\prestamos_biblioteca_100.csv
Validando registros...
Creando tablas (si no existen)...
Limpiando tablas antes de cargar (ejecucion idempotente)...
Cargando dimensiones...
Cargando fact_prestamos...
Registrando errores en etl_errores...
Registrando ejecucion en etl_log...
Generando evidencias\reporte_ejecucion.txt...

--- RESUMEN ---
Filas leidas: 100
Filas cargadas: 98
Filas rechazadas: 2
Estado: FINALIZADO CON ERRORES
PS C:\Users\orlan\OneDrive\Escritorio\Unidad2_ETL_Biblioteca_RuizSantosOrlando>
```

The right sidebar shows the `CHAT` and `SESSIONS` panels. The `CHAT` panel displays a tip: "Tip: Use /create-agent to scaffold a custom agent for your workflow." The `SESSIONS` panel shows a session named `etl_biblioteca.py` with a description "Describe what to build" and a status of "Auto".

3. Captura de las tablas creadas en MySQL:

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'biblioteca_dw' selected. The central SQL editor contains the command 'SHOW TABLES;'. The bottom panel shows the 'Result Grid' with the following tables listed:

Tables_in_biblioteca_dw
dim_alumno
dim_carrera
dim_fecha
dim_libro
dim_sede
eti_errores
eti_log
fact_prestamos

The bottom right panel shows the 'Output' tab with a table of query results:

#	Time	Action	Message	Duration / Fetch
15	11:09:50	SELECT s.sede, AVG(f.dias_prestamo) AS promedio_dias_prestamo FROM fact_prestamos f JOIN dim_se...	4 row(s) returned	0.000 sec / 0.000 sec
16	11:10:30	SELECT l.libro, l.categoria, SUM(f.total_multa) AS total_multa FROM fact_prestamos f JOIN dim_libro l ON f...	5 row(s) returned	0.000 sec / 0.000 sec
17	11:11:47	SELECT f.id_prestamo, fe.fecha, a.alumno, c.carrera, l.libro, l.categoria, s.sede, f.total...	98 row(s) returned	0.000 sec / 0.000 sec
18	11:12:03	SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s...	4 row(s) returned	0.015 sec / 0.000 sec
19	11:20:38	SHOW TABLES	8 row(s) returned	0.015 sec / 0.000 sec

A warning message on the right side of the SQL editor states: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

4. Captura de `fact\_prestamos` con datos:

MySQL Workbench

Local instance MySQL8...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- alumnos_db
- biblioteca
- biblioteca_dw**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- chat
- empleados
- employee_db
- empresa
- hospitallando
- hospitalsaludvital
- linkedin_data
- mascofata
- mi_sistema_tutores
- perros
- productos_db
- rac_db
- ris
- sakila
- sftv
- sys
- tienda
- tricks
- vehiculos

Administration Schemas

Information

Schema: biblioteca_dw

SQL File 13* pacientes tratamientos tratamientos SQL File 6* SQL File 7* SQL File 8*

Limit to 1000 rows

1 * `SELECT * FROM fact_prestamos;`

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

	id_prestamo	id_fecha	id_alumno	id_carrera	id_libro	id_sede	dias_prestamo	multa_diaria	total_multa
▶	5001	1	2	2	1	1	3	0.00	0.00
	5002	2	10	4	2	3	5	5.00	25.00
	5003	3	19	3	3	4	7	10.00	70.00
	5004	4	16	5	4	2	2	15.00	30.00
	5005	5	9	1	5	1	4	0.00	0.00
	5006	6	3	2	6	3	6	5.00	30.00
	5007	7	11	4	7	4	8	10.00	80.00
	5008	8	8	3	8	2	1	15.00	15.00
	5009	9	4	5	9	1	9	0.00	0.00
	5010	10	15	1	10	3	10	5.00	50.00
	5011	11	18	2	1	4	3	10.00	30.00
	5012	12	1	4	2	2	5	15.00	75.00
	5013	13	6	3	3	1	7	0.00	0.00
	5014	14	12	5	4	3	2	5.00	10.00
	5015	15	5	1	5	4	4	10.00	40.00
	5016	16	17	2	6	2	6	15.00	90.00

fact_prestamos 9 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
16	11:10:30	SELECT l.libro, l.categoria, SUM(f.total_multa) AS total_multa FROM fact_prestamos f JOIN dim_libro l ON f...	5 row(s) returned	0.000 sec / 0.000 sec
17	11:11:47	SELECT f.id_prestamo, fe.fecha, a.alumno, c.carrera, l.libro, l.categoria, s.sede, f.total...	98 row(s) returned	0.000 sec / 0.000 sec
18	11:12:03	SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s...	4 row(s) returned	0.015 sec / 0.000 sec
19	11:20:38	SHOW TABLES	8 row(s) returned	0.015 sec / 0.000 sec
20	11:22:01	SELECT * FROM fact_prestamos LIMIT 0, 1000	98 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

11:22 a. m.
05/07/2026

5. Captura de `etl\_errores` con los 2 errores:

MySQL Workbench

Local instance MySQL8... x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- alumnos_db
- biblioteca
- biblioteca_dw**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- chat
- empleados
- employee_db
- empresa
- hospiorlando
- hospitalsaludvital
- linkedin_data
- mascotas
- mi_sistema_tutores
- perros
- productos_db
- rac_db
- ris
- sakila
- sfti
- sys
- tienda
- tricks
- vehiculos

Administration Schemas

Information

Schema: biblioteca_dw

Object Info Session

SQL File 13* pacientes tratamientos tratamientos SQL File 6* SQL File 7* SQL File 8* x

Limit to 1000 rows

```
1 SELECT * FROM etl_errores;
```

Result Grid

#	id_error	fecha_error	archivo_origen	fila_csv	id_registro	descripcion_error	datos_originales
1	1	2026-07-05 11:19:38	prestamos_biblioteca_100.csv	101	5002	id_prestamo duplicado	{"id_prestamo": 5002, "fecha_presta...
2	2	2026-07-05 11:19:38	prestamos_biblioteca_100.csv	100	5099	total_multa incorrecto (esperado=70...	{"id_prestamo": 5099, "fecha_presta...

etl_errores 10 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	11:11:47	SELECT f.id_prestamo, fe.fecha, a.alumno, c.carrera, l.libro, l.categoria, s.sede, f.total...	98 row(s) returned	0.000 sec / 0.000 sec
18	11:12:03	SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s...	4 row(s) returned	0.015 sec / 0.000 sec
19	11:20:38	SHOW TABLES	8 row(s) returned	0.015 sec / 0.000 sec
20	11:22:01	SELECT * FROM fact_prestamos LIMIT 0, 1000	98 row(s) returned	0.000 sec / 0.000 sec
21	11:22:49	SELECT * FROM etl_errores LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Apply Revert Context Help Snippets

11:22 a. m.
05/07/2026

6. Captura de `etl\_log`:

MySQL Workbench

Local instance MySQL8...

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- alumnos_db
- biblioteca
- biblioteca_dw**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- chat
- empleados
- employee_db
- empresa
- hospiorlando
- hospitalsaludvital
- linkedin_data
- mascotas
- mi_sistema_tutores
- perros
- productos_db
- rac_db
- ris
- sakila
- sfti
- sys
- tienda
- tricks
- vehiculos

Administration Schemas

Information

Schema: biblioteca_dw

Object Info Session

SQL File 13* pacientes tratamientos tratamientos SQL File 6* SQL File 7* SQL File 8*

Limit to 1000 rows

```
1 • SELECT * FROM etl_log;
```

Result Grid

	id_log	fecha_ejecucion	archivo_origen	filas_leidas	filas_cargadas	filas_rechazadas	estado
▶	1	2026-07-05 10:43:37	prestamos_biblioteca_100.csv	100	98	2	FINALIZADO_CON_ERRORES
	2	2026-07-05 10:45:45	prestamos_biblioteca_100.csv	100	98	2	FINALIZADO_CON_ERRORES
	3	2026-07-05 11:05:17	prestamos_biblioteca_100.csv	100	98	2	FINALIZADO_CON_ERRORES
	4	2026-07-05 11:19:38	prestamos_biblioteca_100.csv	100	98	2	FINALIZADO_CON_ERRORES
*	MAX	MAX	MAX	MAX	MAX	MAX	MAX

etl_log 11 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	11:12:03	SELECT s.sede, COUNT(*) AS total_prestamos FROM fact_prestamos f JOIN dim_sede s ON f.id_sede = s...	4 row(s) returned	0.015 sec / 0.000 sec
19	11:20:38	SHOW TABLES	8 row(s) returned	0.015 sec / 0.000 sec
20	11:22:01	SELECT * FROM fact_prestamos LIMIT 0, 1000	98 row(s) returned	0.000 sec / 0.000 sec
21	11:22:49	SELECT * FROM etl_errores LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
22	11:23:30	SELECT * FROM etl_log LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

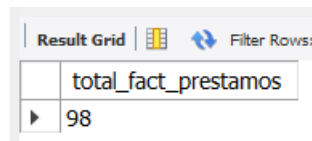
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Apply Revert Context Help Snippets

11:23 a. m.
05/07/2026

7. Captura o evidencia de las consultas SQL:

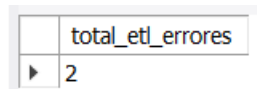
1. Cuantos registros hay en `fact\_prestamos`.



The screenshot shows a 'Result Grid' with a single row containing the value 98. Above the grid, there is a 'Filter Rows' button and a small icon. The query 'total_fact_prestamos' is visible in the header of the grid.

	total_fact_prestamos
▶	98

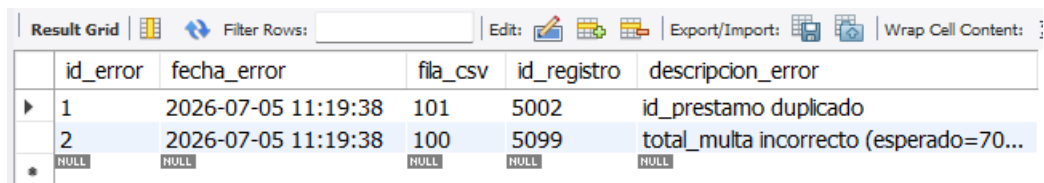
2. Cuantos registros hay en `etl\_errores`.



The screenshot shows a 'Result Grid' with a single row containing the value 2. Above the grid, there is a 'Filter Rows' button and a small icon. The query 'total_etl_errores' is visible in the header of the grid.

	total_etl_errores
▶	2

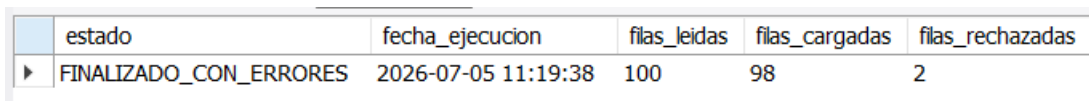
3. Que errores fueron registrados.



The screenshot shows a 'Result Grid' with two rows of error data. The first row has id_error 1, fecha_error 2026-07-05 11:19:38, fila_csv 101, id_registro 5002, and descripcion_error 'id_prestamo duplicado'. The second row has id_error 2, fecha_error 2026-07-05 11:19:38, fila_csv 100, id_registro 5099, and descripcion_error 'total_multa incorrecto (esperado=70...'. There are also some NULL values in the first row.

	id_error	fecha_error	fila_csv	id_registro	descripcion_error
▶	1	2026-07-05 11:19:38	101	5002	id_prestamo duplicado
	2	2026-07-05 11:19:38	100	5099	total_multa incorrecto (esperado=70...

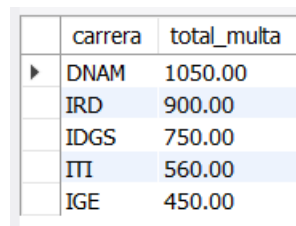
4. Cual fue el último estado registrado en `etl\_log`.



The screenshot shows a 'Result Grid' with one row of data. The state is FINALIZADO_CON_ERRORES, fecha_ejecucion is 2026-07-05 11:19:38, filas_leidas is 100, filas_cargadas is 98, and filas_rechazadas is 2.

	estado	fecha_ejecucion	filas_leidas	filas_cargadas	filas_rechazadas
▶	FINALIZADO_CON_ERRORES	2026-07-05 11:19:38	100	98	2

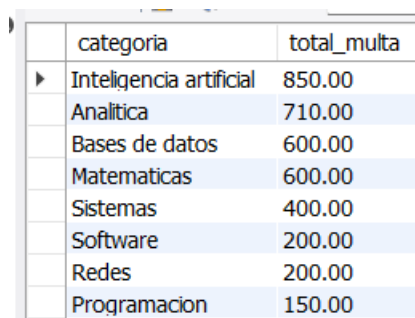
5. Total de multas por carrera.



The screenshot shows a 'Result Grid' with five rows of data. The careers and their total fines are: DNAM (1050.00), IRD (900.00), IDGS (750.00), ITI (560.00), and IGE (450.00).

	carrera	total_multa
▶	DNAM	1050.00
	IRD	900.00
	IDGS	750.00
	ITI	560.00
	IGE	450.00

6. Total de multas por categoria de libro.



The screenshot shows a 'Result Grid' with eight rows of data. The book categories and their total fines are: Inteligencia artificial (850.00), Analitica (710.00), Bases de datos (600.00), Matematicas (600.00), Sistemas (400.00), Software (200.00), Redes (200.00), and Programacion (150.00).

	categoria	total_multa
▶	Inteligencia artificial	850.00
	Analitica	710.00
	Bases de datos	600.00
	Matematicas	600.00
	Sistemas	400.00
	Software	200.00
	Redes	200.00
	Programacion	150.00

7. Promedio de dias de prestamo por sede.

	sede	promedio_dias_prestamo
►	Biblioteca Central	6.2000
	Sala Sur	6.0833
	Sala Norte	4.8000
	Laboratorio Digital	4.5833

8. Los 5 libros con mayor total de multa.

	libro	categoria	total_multa
►	Fundamentos de IA	Inteligencia artificial	850.00
	Estadística Aplicada	Matematicas	600.00
	Bases de Datos I	Bases de datos	500.00
	Linux para Todos	Sistemas	400.00
	Análisis de Datos	Analítica	360.00

9. Prestamos detallados con fecha, alumno, carrera, libro, categoria, sede y total de multa.

id_prestamo	fecha	alumno	carrera	libro	categoria	sede	total_multa
5029	2026-06-01	Daniel Torres	ITI	Análisis de Datos	Analítica	Biblioteca Central	0.00
5001	2026-06-01	Ana Lopez	IDGS	Python Básico	Programación	Biblioteca Central	0.00
5057	2026-06-01	Valeria Cruz	IRD	Linux para Todos	Sistemas	Biblioteca Central	0.00
5085	2026-06-01	Laura Diaz	DNAM	Ingeniería de Software	Software	Biblioteca Central	0.00
5086	2026-06-02	Carlos Luna	IDGS	Estadística Aplicada	Matematicas	Sala Norte	30.00
5030	2026-06-02	Paola Reyes	DNAM	Fundamentos de IA	Inteligencia artificial	Sala Norte	50.00
5002	2026-06-02	Luis Perez	IRD	Bases de Datos I	Bases de datos	Sala Norte	25.00
5058	2026-06-02	Hector Medina	IGE	SQL Practico	Bases de datos	Sala Norte	5.00
5059	2026-06-03	Natalia Flores	ITI	Análisis de Datos	Analítica	Sala Sur	90.00
5003	2026-06-03	Sofía Ruiz	IGE	Minería de Datos	Analítica	Sala Sur	70.00
5031	2026-06-03	Roberto Silva	IDGS	Python Básico	Programación	Sala Sur	30.00
5087	2026-06-03	Maria Solís	IRD	Linux para Todos	Sistemas	Sala Sur	80.00
5060	2026-06-04	Oscar Herrera	DNAM	Fundamentos de IA	Inteligencia artificial	Laboratorio Digital	150.00
5088	2026-06-04	Jorge Cano	IGE	SQL Practico	Bases de datos	Laboratorio Digital	15.00
5004	2026-06-04	Pedro Mata	ITI	Redes de Computad...	Redes	Laboratorio Digital	30.00
5032	2026-06-04	Alejandra Mora	IRD	Bases de Datos I	Bases de datos	Laboratorio Digital	75.00
5033	2026-06-05	Fernanda Rios	IGE	Minería de Datos	Analítica	Biblioteca Central	0.00
5005	2026-06-05	Laura Diaz	DNAM	Ingeniería de Software	Software	Biblioteca Central	0.00
5061	2026-06-05	Ana Lopez	IDGS	Python Básico	Programación	Biblioteca Central	0.00
5089	2026-06-05	Daniel Torres	ITI	Análisis de Datos	Analítica	Biblioteca Central	0.00

(hay más datos abajo, pero solo coloque captura de los primeros)

10. Conteo de prestamos por sede.

	sede	total_prestamos
►	Biblioteca Central	25
	Sala Norte	25
	Laboratorio Digital	24
	Sala Sur	24